Module 4: Higher-Order Linear ODEs (Constant Coefficients)

Welcome to **Module 4**. In the previous modules, we dealt with first-order equations where the highest derivative was $\frac{dy}{dx}$. Now, we advance to **Higher-Order Linear Differential Equations**, which involve second derivatives ($\frac{d^2y}{dx^2}$), third derivatives, and beyond.

These equations are fundamental in analyzing mechanical vibrations, electrical circuits (RLC circuits), and structural engineering. This module focuses specifically on linear equations with **Constant Coefficients**.

Module Learning Objectives

By the end of this module, you will be able to: 1. Understand the structure of the General Solution: $y = y_c + y_p$. 2. Solve homogeneous linear equations by finding the roots of the Characteristic Equation (to find y_c). 3. Apply specific methods to find the Particular Integral (y_p) for different types of non-homogeneous functions (Exponential, Algebraic, Trigonometric).

Topic Index & Solution Methods

The solution to a non-homogeneous linear ODE $f(D)y = F(x)$ is always the sum of two parts:

$$y = y_c + y_p$$

Navigate through the methods for finding these parts using the links below:

1. Finding the Complementary Function (y_c)

- **The Concept:** Solving the homogeneous part $f(D)y = 0$.
- **The Method:** Using the Auxiliary (Characteristic) Equation to find roots.
- **Cases:** Real & Distinct, Real & Repeated, and Complex Conjugate roots.

2. Particular Integral (y_p): Exponential Case

- **Function:** $F(x) = e^{ax}$
- **Rule:** Replace D with a.
- *Includes the "Case of Failure" where $f(a) = 0$.*

3. Particular Integral (y_p): Algebraic Case

- **Function:** $F(x) = x^n$ (Polynomials)
- **Rule:** Expand $[f(D)]^{-1}$ using the Binomial Theorem.
- *Includes step-by-step solved questions.*

4. Particular Integral (y_p): Trigonometric Case

- **Function:** $F(x) = \sin(ax)$ or $\cos(ax)$
- **Rule:** Replace D^2 with $-a^2$.
- *Includes the "Case of Failure" handling.*

5. Particular Integral (y_p): Shift & Product Rules

- **Function:** $F(x) = e^{ax}V(x)$ (Product of exponential and another function)
- **Rule:** The Exponential Shift Theorem.
- *Includes solved examples combining multiple methods.*

Next Steps: The first step in solving *any* higher-order linear equation is finding the Complementary Function. Proceed to Topic 1: Finding y_c.